

A histogram

A histogram groups numerical values into intervals and displays their frequencies.

How does its horizontal axis differ from a categorical bar graph?

Define the intervals

Synthetic download times: 2, 4, 5, 7, 9, 10, 13, 18 seconds. Use intervals $[0,5)$, $[5,10)$, $[10,15)$, $[15,20)$.

Where does a time of exactly 5 seconds belong?

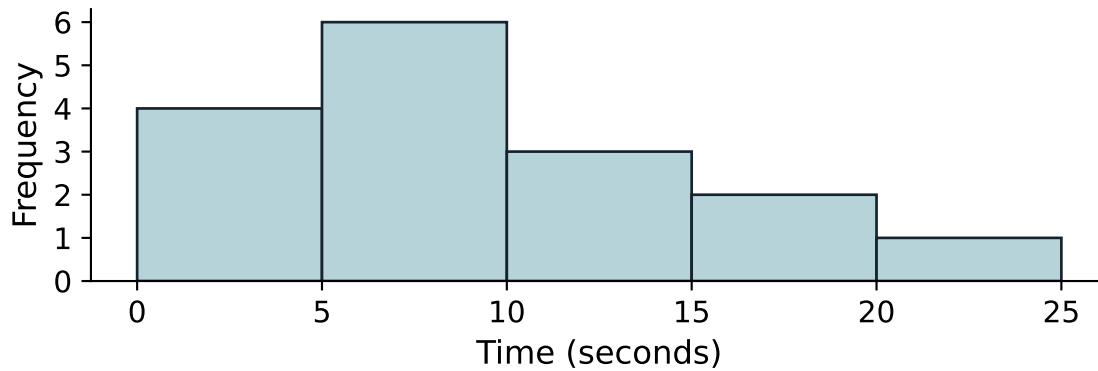
Count the bins

Synthetic download times: 2, 4, 5, 7, 9, 10, 13, 18 seconds. Use intervals $[0,5)$, $[5,10)$, $[10,15)$, $[15,20)$.

What frequency belongs in each interval?

Read a histogram

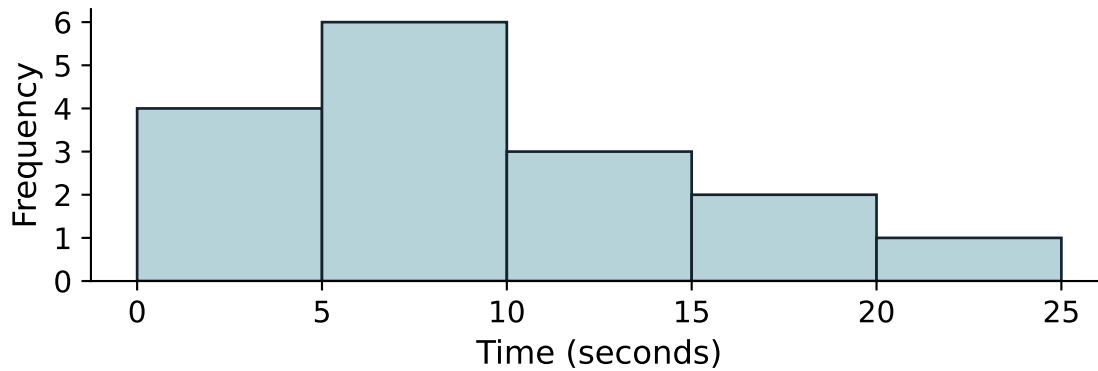
Synthetic times; bins are left-inclusive.



How many downloads took less than 10 seconds?

Read an interval

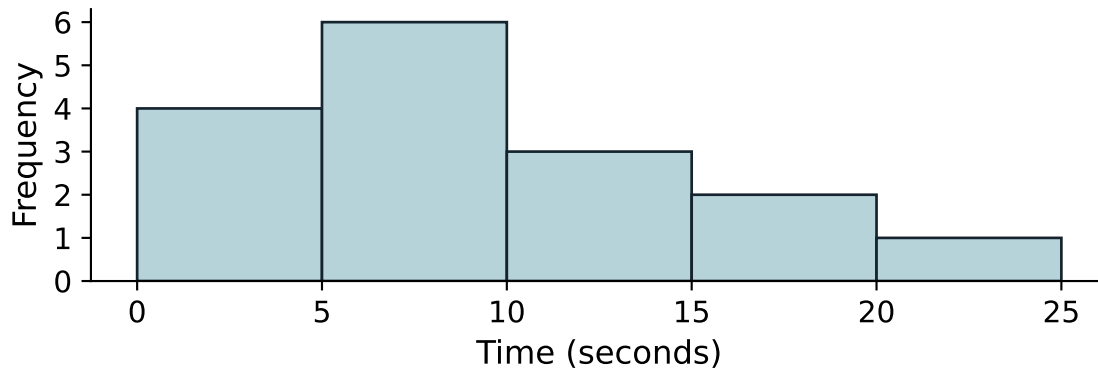
Synthetic times; bins are left-inclusive.



Which five-second interval contains the most observations?

Frequency to percentage

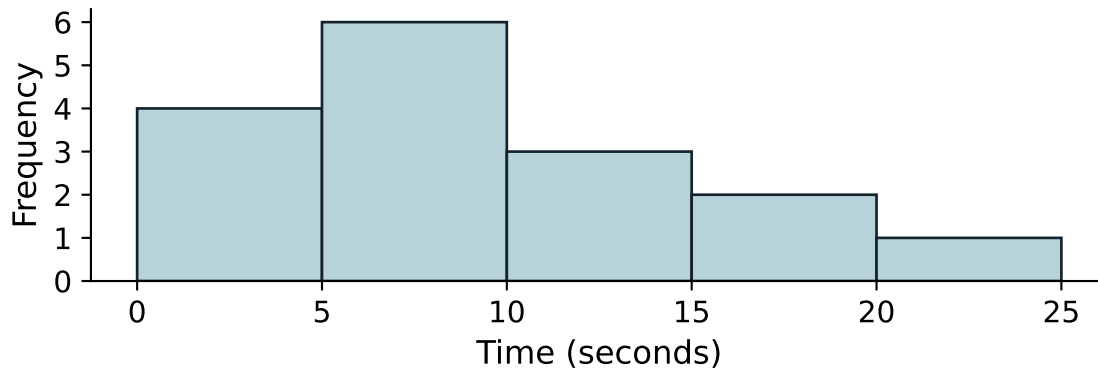
Synthetic times; bins are left-inclusive.



What percentage took from 10 to less than 15 seconds?

Information hidden by grouping

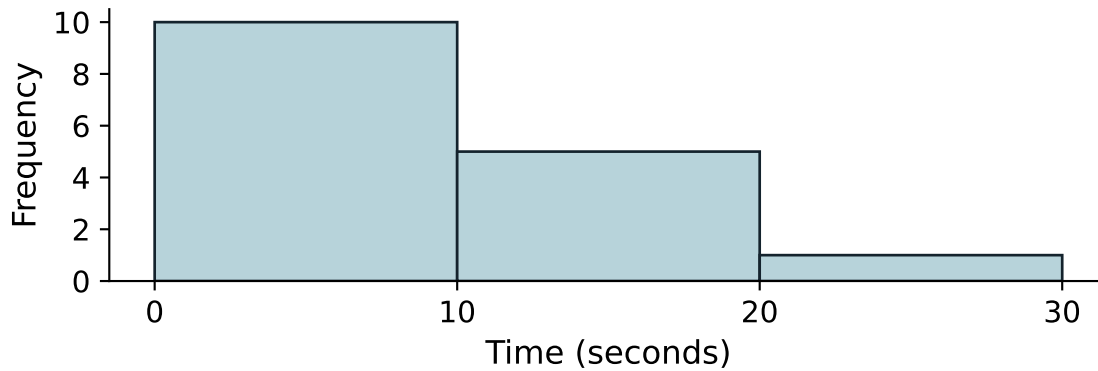
Synthetic times; bins are left-inclusive.



Can this graph reveal the exact number of times equal to 8 seconds?

Bin width

Synthetic times; bins are left-inclusive.



How does a wider bin change the detail we can see?

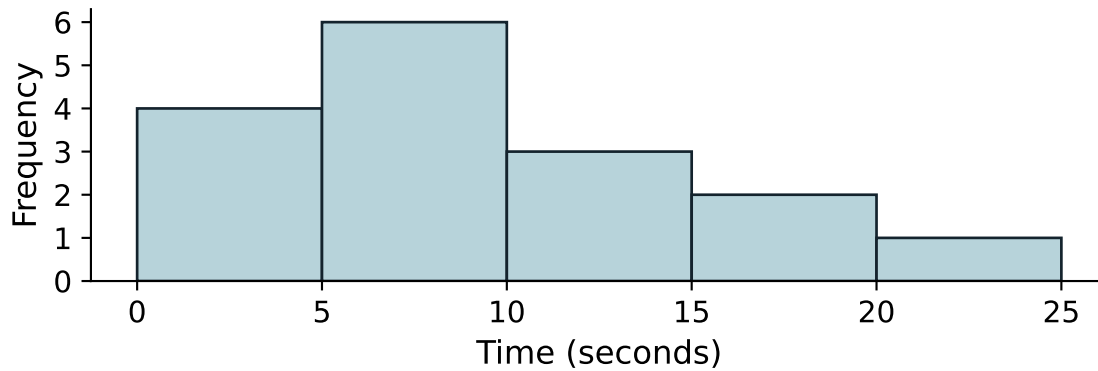
A gap

Synthetic file sizes (KB): 11, 12, 14, 16, 31, 33, 35.

What will a histogram with bins $[10,20)$, $[20,30)$, $[30,40)$ reveal?

Describe the shape

Synthetic times; bins are left-inclusive.



Is the distribution roughly symmetric or longer-tailed on one side?

Label the axes

A histogram has bars of heights 4, 7, and 9 but no axis labels.

What must we know before interpreting these bars?

Relative-frequency histogram

Synthetic intervals have counts 4, 7, 9 out of 20 files.

What heights would a relative-frequency histogram use?

A frequency polygon

A frequency polygon joins frequencies plotted at class midpoints.

For intervals $[0,5)$, $[5,10)$, $[10,15)$, where are the midpoints?

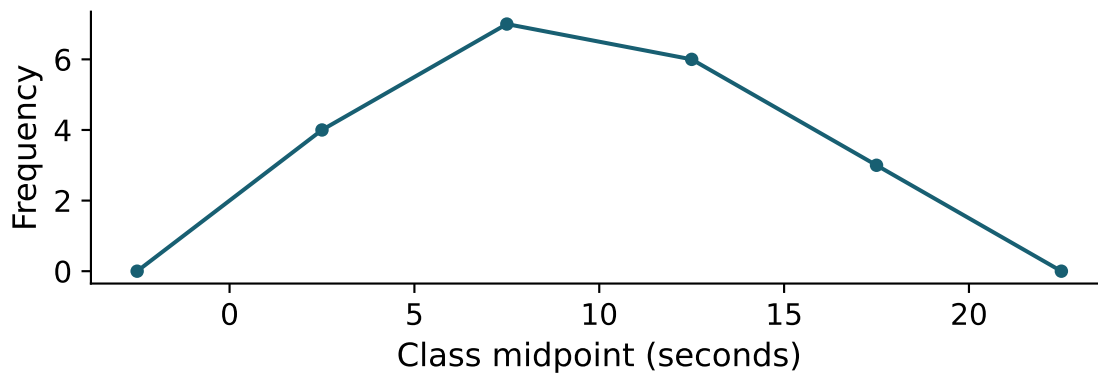
Build a frequency polygon

Synthetic download times. Equal-width class midpoints are 2.5, 7.5, 12.5, 17.5; counts are 4, 7, 6, 3.

What points should the polygon connect?

Read a frequency polygon

Synthetic grouped times; zero endpoints close the polygon.



Which class has the greatest frequency?

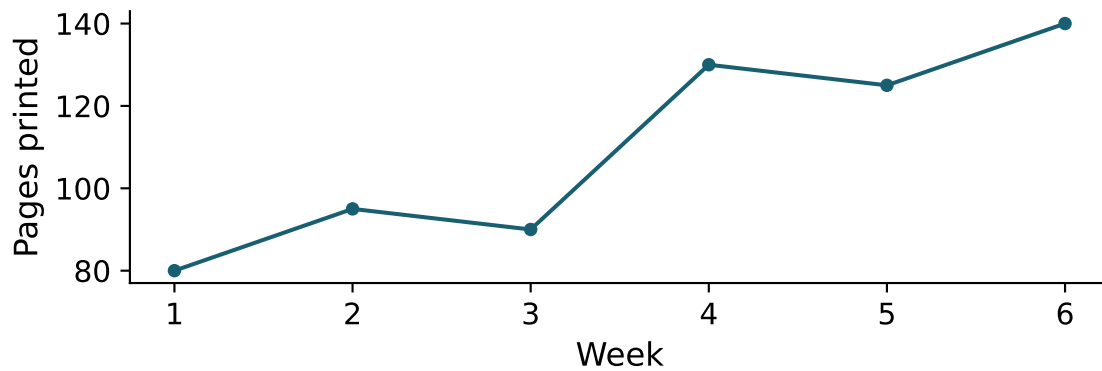
A time series

A time series records a variable in chronological order.

Why must the horizontal axis preserve time order?

Read a time series

Synthetic weekly print totals.



During which adjacent weeks did printing increase most?

Distribution or time order?

Synthetic weekly print totals: 80, 95, 90, 130, 125, 140.

What information would a histogram lose compared with a time-series graph?

Unequal widths

One histogram bin spans 0–5 seconds; another spans 5–20 seconds. Both contain 10 observations.

Why might equal bar heights misrepresent the concentration of data?

Choose a graph

We record file sizes and the total files downloaded on each of 30 consecutive days.

Which graph would best show the change in daily totals?